Time series analysis and forecasting of Austrian energy market prices using ARIMA/ARIMAX modeling

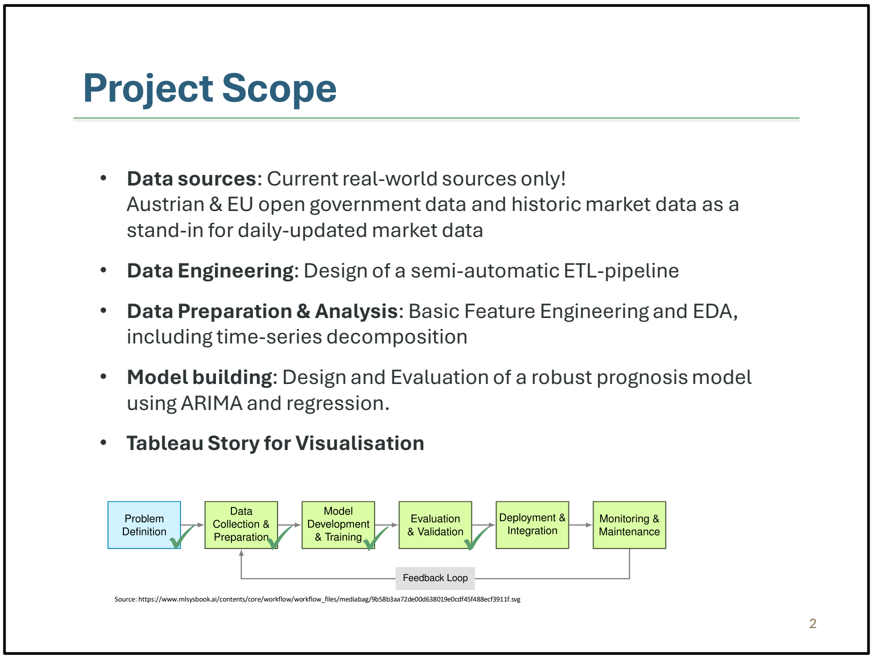
Capstone Project for the 2025 „Data Practictioner“ Class at neuefische

Paul Ringler | 27.10.2025

Business Case

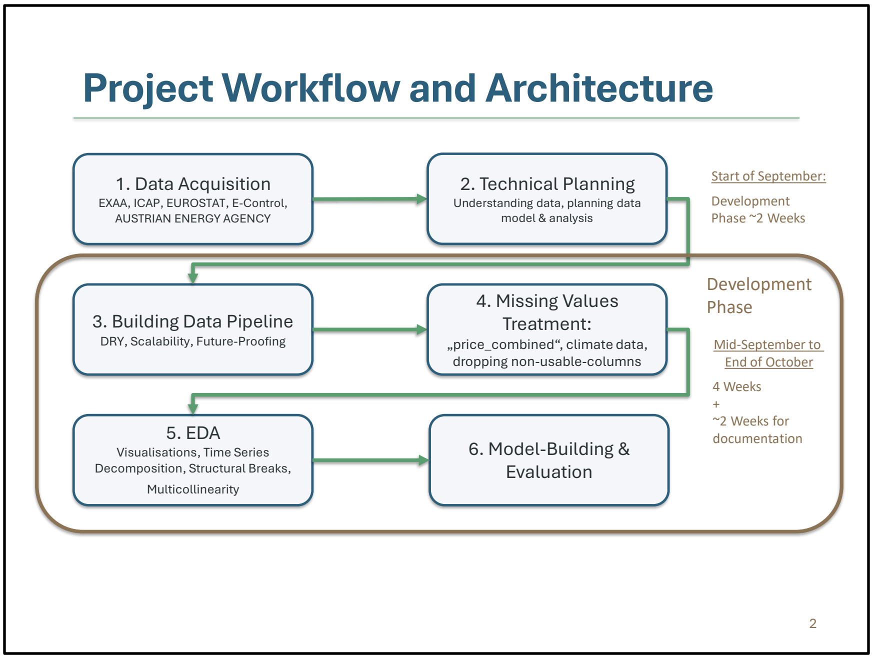
- Following the russian full-scale invasion of Ukraine in 02/2022, Austrian energy prices have experienced significant increases and have also become very volatile.
- Large power producers (e.g. wind/solar parks, public utility companies) and power consumers (e.g. manufacturing companies, data centers need planning security.
- This best achieved by securing over-the-counter supply contracts. These are sold at fixed prices for different durations (e.g. 1-6 month-ahead).
- Forecasting these prices yields benefits in terms of risk-management as well as cost optimization in procurement planning.
- This project explores the cornerstones of a light-weight, continuous forecasting application designed for use by small-scale manufacturing and power supply companies in Austria

- Post-Ukraine invasion: Austrian energy prices surged and became highly volatile
- - Target audience: Small manufacturing companies, power suppliers
- - Over-the-counter contracts: fixed prices for 1-6 months ahead
- - Forecasting enables: (1) Risk hedging, (2) Optimal contract timing
- - This project: Lightweight forecasting tool prototype using ARIMA/ARIMAX



- Classic ML workflow: Problem Definition → Data → Model → Evaluation
- This project covers: ✓ Data Engineering, ✓ EDA, ✓ Model Building, ✓ Evaluation
- Out of scope: Deployment, Monitoring (future work)
- 6 data sources: Electricity prices, CO2, Gas, Climate, Production Mix, Economy
- Semi-automated ETL pipeline designed for continuous updates. I.e: Data extraction, then transformation and finally shaping into a data file for further work. Code once, and then analyse.
- (vs. loading the data into a raw file and only transforming it whenever it is needed for analysis) Transformation Code for each analysis.

In this presentation I´ll start with an overview of the data and crucial EDA findings, then present the results of model building and the final model.
I also have a slide with details on the ETL-pipeline which I´ll gladly go through, if there is enough time.



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Data Overview

Data Granularity: Monthly

- **Outcome Variable:** Average of day-ahead electricity prices (EXAA / Vienna Stock Exchange)
- **Input Variables**:
    - Gas prices - Average of 1-month ahead futures (Austrian Energy Agency)
    - Price of carbon emission credits in European Trading Area (ICAP)
    - Climate data for austria (EUROSTAT)
    - Proportion of renewables in energy production (E-Control)
    - Economic Indicators (STATISTIK Austria)

Central EDA Finding

Three distinct periods in Data:

Period	~Timespan	No. of months	Mean (EUR/MWh)	SD	Stationary?
Pre-Shock	2015-2020	72	35,94	9,6	Yes (ADF p= ,034)
Price Shock	2021-2023	26	181,92	112,44	No
Post-Shock	2023-2025	30	22,70	22,7	Yes (ADF p= ,027)

Why is stationarity relevant?

- Stationarity = Constant mean, variance, autocorrelation structure over time
- ARIMA models need stable patterns from the past
- Stationarity was detected using forwards/backwards expanding ADF and KPSS tests

Solution: Train on Price-Shock + Post-Shock (N=56)

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- TO DO: Research, Stationarity Autocorrelation more
- Research ADF and KPSS tests again, short explanation for those

- Three regimes: Stable-Volatile-Stable sandwich structure
- Pre-shock: Low prices, low volatility, stationary
- Price shock: Mean jumped 5x, volatility exploded, non-stationary
- Post-shock: Prices normalized at higher level, volatility decreased, stationary again

- Modeling challenges:
    - Training on full period = learning from regime no longer valid
    - ARIMA requires N=XX months training data minimum and XX for testing for stable model, actual stationary period covers only XX months (so far)
    - Solution: Train on Period (2020-2023) = XX months with available exogenous data

	n_months	mean	median	std
cv	min	max	iqr	

stationary_pre	72.0	35.94	33.50	9.60	0.27	18.0	62.0	10.00
price_shock	26.0	181.92	174.25	112.44	0.62	50.5	488.0	140.25
stationary_post	30.0	89.27	85.00	22.70	0.25	59.0	141.0	29.00

Evaluation of Final Model

ARIMAX (0,0,0) + Indexed Gas Price

Variable	Coef	Std Err	Beta	Significant?
Constant	43,84	4,33	-	No (p = ,323)
Gas price	1,61	^0,41	0,83	Yes (p = < ,001)

r^2 = 0,695, Adj. r^2 = 0,688

- RMSE: 21.14 EUR/MWh (~20-25% error at mean price of 90 EUR/MWh)
  vs. RMSE 30.76 (Persistence) (~XXX% Error)

- Ljung-Box (lag 12): p = 0.83 & (lag 24): p = 0.99 → No autocorrelation
- Shapiro-Wilk: p < 0.001 → Non-normal residuals
- Heteroskedasticity: H = 10.05, p < 0.01

Limitations: Forecast degrades quickly, assumption of stable relationship between electricity and gas prices is brittle. Extreme outliers underestimated. Dependence on outside data.

Needs constant monitoring. Retraining & rebuild on pure post-shock data in XX months

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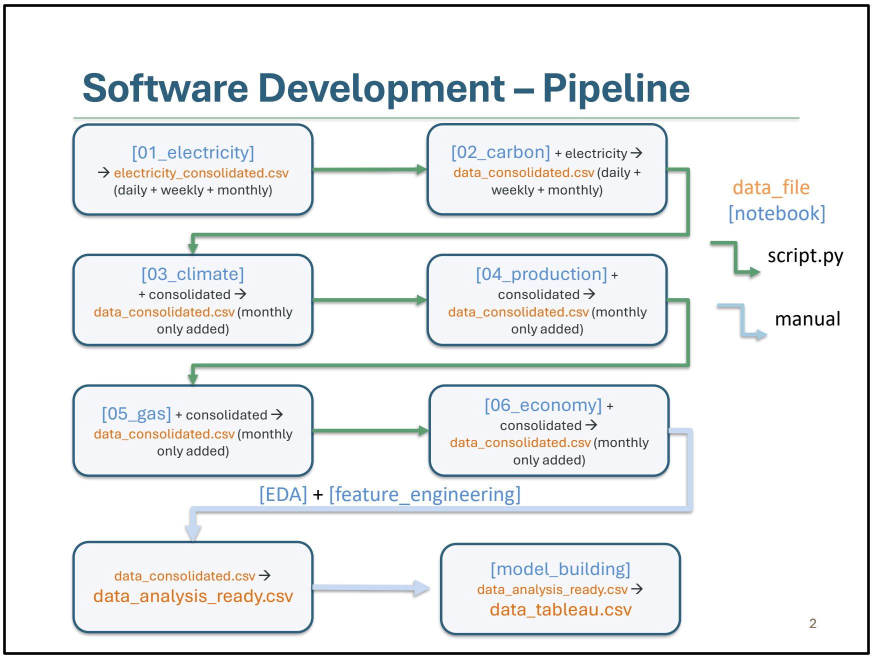
- Ljung-Box tests: Residuals are uncorrelated → Model captured all predictable patterns
- Shapiro-Wilk failure: Residuals have heavy tails → Extreme events harder to predict
- Heteroskedasticity: Error variance not constant → CI widths are approximate
- Bootstrap CI: Realistic uncertainty quantification → 95% CI = ±42 EUR/MWh = ±40-50% of typical price → Still useful for scenario planning, not point predictions


Software Development – Pipeline
[01_electricity]
→ electricity_consolidated.csv
(daily + weekly + monthly)
[02_carbon] + electricity →
data_consolidated.csv (daily +
weekly + monthly)
data_file
[notebook]
script.py
manual
[03_climate]
+ consolidated →
data_consolidated.csv (monthly
only added)
[04_production] +
consolidated →
data_consolidated.csv (monthly
only added)
[05_gas] + consolidated →
data_consolidated.csv (monthly
only added)
[06_economy] +
consolidated →
data_consolidated.csv (monthly
only added)
[EDA] + [feature_engineering]
data_consolidated.csv →
data_analysis_ready.csv
[model_building]
data_analysis_ready.csv →
data_tableau.csv
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DRY – Principle Implementation

TO DO

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Thank you!

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https://github.com/Pringlebot/austrian_electricity_prices_capstone/tree/main